

EM-seq Method

1. DNA Extraction and Quantification

After DNA were extracted from the samples (Animals: TIANGEN DP705; Plants: TIANGEN DP342; cfDNA: QIAamp MinElute cfDNA Midi Kit (Qiagen 55284)), Genomic DNA degradation and contamination was checked using agarose gels. DNA purity was evaluated by NanoPhotometer® spectrophotometer (IMPLEN, CA, USA). Fragment distribution of cfDNA or Genomic DNA was checked using Agilent 2100. DNA concentration was measured using Qubit® DNA Assay Kit in Qubit® 2.0 Fluorometer (Life Technologies, CA, USA)

2. Library Preparation and Quantification

Unmethylated Lambda DNA was spiked into each sample (10-100 ng total DNA input) as a negative control for methylation analysis. The DNA was then sheared to 200-350 bp fragments using the Bioruptor Pico ultrasonicator. Library preparation was performed using the Enzymatic Methylation sequencing Kit. Final library quality and fragment size distribution were assessed using the Agilent Bioanalyzer 2100 system.

3. Sequencing

The library was sequenced on an Illumina platform (Illumina, CA, USA) to generate 150 bp paired-end reads. Image analysis and base calling were performed using the Illumina CASAVA pipeline.

4. Data Analysis

4.1 Quality Control

Quality statistics of the raw reads was evaluated using FastQC (fastqc_v0.11.3). Raw reads in FASTQ format were pre-processed through fastp (fastp 0.20.0). The remaining reads after filtering were proceeded to subsequent analyses. The quality statistics of clean reads was also performed using FastQC.

4.2 Reference Data Preparation before Analysis

Prior to conducting the analysis, we obtained or prepared the necessary reference datasets for the studied species. This included the reference genome sequence in FASTA format, gene annotation in GTF format, Gene Ontology (GO) annotation files, functional description files, and a BED file specifying gene regions. Repetitive elements in the genome were identified and masked using RepeatMasker, which screens for interspersed repeats and low-complexity DNA sequences by referencing curated repeat libraries such as Dfam or RepBase. Additionally, CpG islands (CGIs) were predicted from the genome sequence using the cpGIslandExt tool, generating a track of these genomic features for downstream analysis.

4.3 Reads Mapping to the Reference Genome

The alignment of bisulfite-treated reads to the reference genome was performed using Bismark software (version 0.16.3) with the parameter -X 700 to set the maximum fragment length for valid paired-end alignments and --dovetail to allow dovetailed pairs[1]. Prior to alignment, the reference genome was in silico converted into bisulfite-converted versions: one with all cytosines (C) transformed to thymines (T) and another with all guanines (G) transformed to adenines (A). These converted genomes were indexed using Bowtie2 to facilitate efficient alignment[2]. Clean sequencing reads were similarly converted into fully bisulfite-transformed versions (C-to-T and G-to-A) and aligned directionally to the converted reference genomes. For each read, alignments to both the original top strand and the complementary bottom strand were evaluated. Reads producing a unique best alignment across these processes were retained. The methylation state of each cytosine position was inferred by comparing these alignments to the unconverted genomic sequence.

PCR duplicates were identified as reads aligning to identical genomic regions and removed using Bismark's deduplication function to ensure accurate methylation calling. Sequencing depth and coverage statistics were calculated based on deduplicated reads. Methylation calls were extracted using "bismark_methylation_extractor with the --no_overlap" option to avoid double-counting overlapping read pairs in paired-end data. The results were converted into bigWig format for visualization in the IGV browser . To assess bisulfite conversion efficiency, the non-conversion rate was calculated as the

percentage of cytosines remaining unconverted (i.e., still sequenced as cytosine) at reference cytosine positions in the lambda phage genome (spike-in control), providing a quality control metric for the bisulfite treatment step

4.4 Estimating Methylation Level

Methylated sites were identified using a binomial test based on methylated read counts (mC), total read counts (mC + umC), and the non-conversion rate (r). Sites with a methylation level (ML) ≥ 0.05 and an FDR-adjusted p-value < 0.01 were considered significantly methylated[3]. To analyze methylation across the sequence, the genome was divided into 10 kb bins. For each window, the total methylated and unmethylated counts were summed. The methylation level for each window or cytosine site was calculated as the proportion of methylated cytosines, defined as $ML = mC / (mC + umC)$.

$$ML(C) = \frac{reads(mC)}{reads(mC) + reads(C)}$$

4.5 Differentially Methylated Analysis

Differentially methylated regions (DMRs) were identified using the DSS software[4-6]. The core of DSS is a new dispersion shrinkage method for estimating the dispersion parameter from Gamma-Poisson or Beta-Binomial distributions. According to the distribution of DMRs through the genome, we defined the genes related to DMRs as genes whose gene body region (from TSS to TES) or promoter region (2 kb upstream from the TSS) have an overlap with the DMRs.

4.6 GO and KEGG Enrichment Analysis of DMR-related Genes

Gene Ontology (GO) enrichment analysis of genes related to DMRs was implemented by the Goseq R package[7], in which gene length bias was corrected. GO terms with corrected P-value less than 0.05 were considered significantly enriched by DMR-related genes. We used KOBAS software to test the statistical enrichment of DMR-related genes in KEGG pathways[8, 9].

5. References

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